SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BCSE302P - DATABASE SYSTEMS LAB

Semester : Winter Semester 25-26 **Faculty : Dr.S.Geetha**

EXERCISE 3

Title: Implementation of Data Manipulation Language (DML) Commands

1. OBJECTIVE

To understand and implement **Data Manipulation Language (DML) commands** in SQL by performing **INSERT, UPDATE, DELETE**, and **basic SELECT operations** on relational database tables, and to retrieve, modify, and manage data effectively using appropriate conditions.

Example Table :

STUDENT

Column Name	Data Type
ROLLNO	INT
NAME	VARCHAR(30)
DEPT	VARCHAR(20)
MARKS	INT

1. INSERT Command

Purpose

Used to insert new records into a table.

There are 4 different styles of INSERT Command

Style 1: INSERT with VALUES (All Columns)

- **Syntax:**
  INSERT INTO TABLE_NAME VALUES (value1, value2, value3, ..);
- **Example:**
  INSERT INTO STUDENT VALUES (101, 'Anitha', 'CSE', 85);

Style 2: INSERT with Column Names

- **Syntax:**

INSERT INTO TABLE_NAME (COLUMN1, COLUMN2, ...) VALUES (value1, value2, ...);

- **Example:**
  INSERT INTO STUDENT (ROLLNO, NAME, MARKS) VALUES (102, 'Rahul', 78);

Style 3: INSERT Multiple Rows

- **Syntax:**
  INSERT INTO TABLE_NAME
  VALUES
  (value1, value2, ...),
  (value1, value2, ...);
- **Example:**
  INSERT INTO STUDENT
  VALUES
  (103, 'Kavya', 'ECE', 90),
  (104, 'Arun', 'MECH', 72);

Style 4: INSERT with & Operator

- **Syntax:**
  INSERT INTO table_name (column1, column2)
  VALUES (&value1, '&value2');
- **Example:**

  INSERT INTO STUDENT VALUES (&ROLLNO, '&NAME',& MARKS);

  Note : For number data type, only &will precede the column name. For char and varchar data type , the column name is enclosed in a single quotes will come. Eg. '&NAME'

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2. UPDATE Command

Purpose

Used to modify existing records in a table.

Style 1: UPDATE Single Column

- **Syntax:**
  UPDATE TABLE_NAME
  SET COLUMN_NAME = value

WHERE condition;

- **Example:**
  UPDATE STUDENT
  SET MARKS = 88
  WHERE ROLLNO = 102;

Style 2: UPDATE Multiple Columns

- **Syntax:**
  UPDATE TABLE_NAME
  SET COLUMN1 = value1, COLUMN2 = value2
  WHERE condition;
- **Example:**
  UPDATE STUDENT
  SET DEPT = 'AI&DS', MARKS = 92
  WHERE ROLLNO = 103;

Style 3: UPDATE Without WHERE (All Rows)

- **Syntax:**
  UPDATE TABLE_NAME
  SET COLUMN_NAME = value;
- **Example:**
  UPDATE STUDENT
  SET MARKS = MARKS + 5;

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3. **DELETE Command**

Purpose

Used to remove records from a table.

Style 1: DELETE Specific Record

- **Syntax:**
  DELETE FROM TABLE_NAME
  WHERE condition;
- **Example:**
  DELETE FROM STUDENT
  WHERE ROLLNO = 104;

Style 2: DELETE Multiple Records

- **Syntax:**
  DELETE FROM TABLE_NAME
  WHERE condition;
- **Example:**

```
DELETE FROM STUDENT
WHERE MARKS < 40;
```

Style 3: DELETE All Records

⚠️ *Deletes all rows but keeps table structure*

- **Syntax:**
  ```
  DELETE FROM TABLE_NAME;
  ```
- **Example:**
  ```
  DELETE FROM STUDENT;
  ```

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4. SELECT Command (Basic Queries)

Purpose

Used to retrieve data from a table.

Style 1: SELECT All Columns

- **Syntax:**
  ```
  SELECT * FROM TABLE_NAME;
  ```
- **Example:**
  ```
  SELECT * FROM STUDENT;
  ```

Style 2: SELECT Specific Columns

- **Syntax:**
  ```
  SELECT COLUMN1, COLUMN2
  FROM TABLE_NAME;
  ```
- **Example:**
  ```
  SELECT NAME, MARKS FROM STUDENT;
  ```

Style 3: SELECT with WHERE Condition

- **Syntax:**
  ```
  SELECT * FROM TABLE_NAME
  WHERE condition;
  ```
- **Example:**
  ```
  SELECT * FROM STUDENT
  WHERE DEPT = 'CSE';
  ```

Style 4: SELECT with ORDER BY

- **Syntax:**
  ```
  SELECT * FROM TABLE_NAME
  ```

ORDER BY COLUMN_NAME ASC | DESC;

- **Example:**

```
SELECT * FROM STUDENT
ORDER BY MARKS DESC;
```

5. EXPECTED OUTCOMES

By the end of this exercise, students will be able to:

- Insert single and multiple records into database tables using different INSERT styles
- Use substitution variables for dynamic data entry
- Modify existing records using the UPDATE command with conditions
- Delete specific or all records using the DELETE command
- Retrieve data using basic SELECT queries
- Understand the effect of WHERE clause in UPDATE, DELETE, and SELECT operations
- Verify data changes by executing SELECT queries